

Fuzzy Exercise Assistant

Final Project Report

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Fuzzy Sets and Systems II

1. Introduction

Regular physical activity is essential for maintaining physical and mental health. However, many people struggle to determine which types of exercise are most suitable for their personal goals, fitness level, and physical condition. Generic exercise recommendations often fail to consider individual differences, resulting in plans that may be ineffective, overwhelming, or difficult to follow.

To address this problem, we developed a Fuzzy Exercise Assistant, a decision support system that generates exercise category recommendations. The system uses a Fuzzy Cognitive Map (FCM) to model the relationships between user different exercise categories such as light cardio, strength training, HIIT and beginner programs.

The objective of this project is not to replace professional fitness advice, but to provide users with simple and understandable exercise guidance tailored to their needs.

2. Problem Definition and Motivation

Choosing an appropriate exercise program can be difficult, especially for beginners.

During our initial design process interviews, we observed that many users were uncertain about how to start exercising and which exercises best suited their goals and physical condition.

Some users reported that they wanted recommendations rather than generic workout advice. Others expressed uncertainty about factors such as body type, fitness level and exercise goals should influence exercise selection.

Based on observations, we identified the following challenge:

How can we provide simple and exercise category recommendations that consider a users fitness level, body characteristics and exercise goals?

To address this challenge, we designed a fuzzy decision support system capable of handling uncertainty and producing recommendations that are more flexible than traditional rule-based systems.

3. Design Science Research Approach

This project follows the Design Science Research (DSR) methodology, which consists of three main stages :Build, Implement and Evaluate.

3.1 Build

The artifact developed in this project is a Fuzzy Exercise Assistant that recommends exercise types based on user characteristics and fitness goals.

3.2 Implement

The assistant was implemented using:

- Python
- FCMpy
- Fuzzy Cognitive Maps
- HTML/CSS/JavaScript frontend
- Recommendation Engine

- Result logging system

The system collects user inputs, processes them through an FCM model, generates exercise category recommendations.

3.2 Evaluate

The artifact was evaluated through user testing. Participants interacted with the system and completed a questionnaire measuring usefulness, safety and usability.

4. System Design and Input Selection

4.1 System Overview

The Fuzzy Exercise Assistant is a decision support system designed to recommend exercise types based on a user's physical characteristics and fitness goals. The system receives user inputs, processes them through a Fuzzy Cognitive Map (FCM), and generates exercise category recommendations.

The recommendation process consists of three stages:

1. User input collection
2. FCM-based reasoning
3. Exercise recommendation generation

The generated recommendations are presented to the user through a web interface together with a ranking of suitable exercise types.

4.2 Input Selection

During the initial design process, several potential factors were considered, including age, motivation, health conditions, weekly activity level, fitness goals, and body characteristics.

However, to keep the first prototype simple, understandable, and easy to evaluate, we selected four core inputs that have a direct influence on exercise recommendations.

BMI

Body Mass Index (BMI) was selected because body composition can influence the suitability of certain exercise types. Users with a higher BMI may benefit from lower-impact activities that place less stress on joints and reduce injury risk. Therefore, BMI directly influences recommendations such as Light Cardio and Beginner Programs.

Fitness Level

Fitness Level represents the user's exercise experience and physical readiness. This variable helps determine whether advanced exercise types, such as HIIT, are appropriate for the user. Individuals with higher fitness levels are generally more capable of safely performing high-intensity exercise.

Muscle Gain Goal

Muscle Gain Goal reflects the user's desire to increase muscle mass. Since strength training is strongly associated with muscle development, this input plays a significant role in determining strength-based recommendations.

Weight Loss Goal

Weight Loss Goal represents the user's intention to reduce body weight or body fat. This objective influences recommendations related to cardiovascular exercise and HIIT training, which are commonly associated with calorie expenditure and weight management.

4.3 Considered but Excluded Inputs

Several additional variables were considered during the design phase but were excluded from the first version of the prototype.

Variable	Reason for Exclusion
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Age	To keep the prototype simple and focused on core fitness variables.
Motivation Level	Difficult to measure objectively and consistently across users.
Medical Conditions	Outside the scope of the project. Users are instead shown a safety warning recommending consultation with a healthcare professional.
Weekly Activity Level	Closely related to fitness level and therefore omitted to reduce complexity.

These variables may be incorporated in future versions of the system to improve recommendation quality and personalization.

4.4 Outputs

The system generates recommendations for four exercise categories:

- **Light Cardio** – low-impact cardiovascular activities such as walking, cycling, or swimming.
- **Strength Training** – resistance exercises designed to increase muscular strength and muscle mass.
- **HIIT** – High-Intensity Interval Training involving alternating periods of intense activity and recovery.
- **Beginner Program** – foundational exercise programs suitable for inexperienced users.

Each output receives a score between 0 and 1. Higher values indicate stronger recommendations. The exercise category with the highest score is presented as the primary recommendation, while additional high-scoring categories may be suggested as complementary exercise options.

5. Fuzzy Cognitive Map Model and Weight Justification

5.1. FCM Structure

The Fuzzy Cognitive Map consists of 8 nodes divided into two categories:

Input Nodes (4):

- BMI
- Fitness_Lvl
- Goal_MuscleGain
- Goal_WeightLoss

Output Nodes (4):

- Rec_Light_Cardio
- Rec_Strength
- Rec_HIIT
- Rec_Beginner

The FCM uses a directed graph structure where input nodes influence output nodes, but output nodes do not influence each other or feed back to inputs.

5.2. Weight Matrix

The following weight table shows the causal relationship between input and output nodes (weights: [-1,1]):

Input Node	Rec_Light_Cardio	Rec_Strength	Rec_HIIT	Rec_Beginner
BMI	+0.7	+0.2	-0.3	+0.8
Fitness_Lvl	-0.2	+0.7	+0.9	-0.8
Goal_MuscleGain	-0.2	+1.0	+0.2	0.0
Goal_WeightLoss	+0.6	+0.3	+0.8	0.0

5.3. Weight Justification

To establish our weights, we reviewed several papers and attempted to ground our weights in this research while still incorporating our intuition.

BMI

The sources suggest that individuals with higher BMI should engage in lighter cardio activities such as yoga or cycling (Benisek, n.d.) because the person may experience joint or other types of pain as they are carrying more weight. It is recommended to find movements that are not harsh on the body. (Benisek, n.d.) These researches suggest the user does some lighter exercises and with less difficulty. This was the reason we assigned **+0.7** to light cardio recommendation, **+0.8** to beginner exercises recommendation and **-0.3** to HIIT.

While for strength-heavy recommendations, there were different views on its role in treating obesity. However, we trusted the American College of Sports Medicine's public recommendation of at least two sessions of resistance training weekly to also help prevent muscle loss during weight loss (Troutman, 2025). Taking into consideration both views we chose to assign lower positive correlation rather than a negative one (**=0.2**).

FITNESS_LVL

The fitness level weights were filled intuitively as usually someone with a higher level of fitness training is familiar with more difficult exercises and would have lower chances of getting injured. This is the reason why we assigned the weights as in the above table.

GOAL_MUSCLEGAIN

According to many papers, cardiovascular exercise does slow muscle growth when performed concurrently with strength training. (Hickson,1980) This resulted in light cardio having a negative correlation (**-0.2**). Whereas, “HIIT promotes increases in lean mass, maximal strength, early RFD, and lower limb muscle power in healthy older and young individuals.” states Manosalva and others in their research article. (Caparrós-Manosalva et al., 2023)

As the main result in strength training is muscle gain, we assigned a **+1.0** immediately.

Because doing beginner exercises has to do more with body condition and fitness knowledge, both of the goals (muscle gain goal and weight loss goal) were assigned with a **0.0**.

GOAL_WEIGHTLOSS

Research suggests exercises included both in light cardio and HIIT effectively aid weight loss while strength training helps maintain the weight and muscles after weight loss has been achieved. (Laguaite, 2025)

As mentioned above, the goals have no correlation with doing beginner exercises.

5.4. Transfer Function

The **sigmoid function**

$$\frac{1}{1+e^{-x}}$$

was selected as the transfer function because:

- output values are bounded between 0 and 1, which maps naturally to recommendation
- provides smooth and differentiable transitions
- models gradual activation of recommendations rather than binary on/off decisions

6. System Implementation

6.1. Technical Architecture

The Fuzzy Exercise Assistant was implemented using a three-tier architecture:

Frontend Layer:

- HTML/CSS/JavaScript for user interface
- Collects user inputs via form elements (sliders/number inputs)
- Displays recommendation results as ranked exercise categories
- Communicates with backend via REST API

Backend Layer:

- Python Flask API (or equivalent) handling HTTP requests
- FCM simulation engine using custom implementation
- Input validation and normalization
- JSON response generation

Processing Layer:

- Fuzzy Cognitive Map inference engine
- Iterative simulation until convergence (threshold < 0.001)
- Sigmoid transfer function application
- Recommendation scoring and ranking

6.3. Input Processing

BMI Normalization (bmi_to_fuzzy)

This function maps the real BMI to [0,1] range after it gets the user's BMI.

low (18.5): the lowest of normal BMI

high (35.0): highest of normal BMI

```
fuzzy_bmi = max(0, min(1, (bmi - 18.5) / (35.0 - 18.5)))
```

- BMI 18.5 → 0.0 (underweight/normal lower bound)
- BMI 26.75 → 0.5 (overweight threshold)
- BMI 35.0 → 1.0 (obese upper bound)

Slider Normalization (slider_to_fuzzy):

This function maps the 1-10 slider used in the input form to a fuzzy decimal.

```
fuzzy_value = (slider_value - 1) / 9.0
```

- Value 1 → 0.0 (lowest priority)
- Value 5 → 0.444 (medium priority)
- Value 10 → 1.0 (highest priority)

6.4. FCM Simulation Algorithm

The system uses the modified Kosko (mKosko) inference mechanism together with the sigmoid transfer function.

During each iteration, the activation values of concepts are updated according to the weighted causal relationships defined in the weight matrix.

The inference process continues until either:

- convergence threshold is reached
- maximum iteration count is reached

The convergence threshold was set to:

$$\varepsilon = 0.001$$

This ensured stable recommendation outputs while avoiding unnecessary computation.

6.4. Recommendation Generation

After convergence, the final activation values of the recommendation nodes are extracted and sorted in descending order.

The recommendation with the highest activation value is presented as the primary recommendation.

Example output:

Recommendation	Score
Light Cardio	0.91
Beginner Program	0.77
HIIT	0.69
Strength Training	0.65

The system therefore recommends Light Cardio as the most suitable exercise category for the given user profile.

6.6. Result Logging and Evaluation Support

To support evaluation and user testing, the backend automatically stores recommendation results in a CSV file.

The logged data includes:

- timestamp
- BMI
- fitness level
- exercise goals
- recommendation scores
- best recommendation category

This logging mechanism enabled later analysis of recommendation patterns and user testing results.

7. Evaluation Methodology

This is the most challenging aspect of our work, as it is difficult to objectively evaluate whether an exercise is good for a specific individual. Instead, we must assess whether the user feels the recommendations are appropriate for them. Among expert evaluation and user evaluation, we decided that user evaluation is more feasible for our prototype. We initially wanted the users to follow the recommendations, record their process and evaluate the recommendations via a questionnaire. However, getting the users to do the exercise and record them was inefficient and not simple to record. That is why we asked questions about the applications themselves and the users' opinions. The participants rated the following statements on a 1-5 Likert scale, all being required (1-strongly disagree, 5-strongly agree):

- The recommended exercises matched my fitness goals.
- The recommendations felt personalized and not generic.
- The recommendations seemed safe for me.

- The recommendations were useful for planning my workouts.
- The application was easy to use.
- I would use this assistant again.
- The recommendations made sense given the information I entered.

and answered 2 open-ended questions:

- What would you improve in the assistant?
- Any other comments?

The users first used the application and entered their information. After the test of the application, they filled out the survey that we had sent the link to and answered the questions.

More on evaluation can be found in our method evaluation documentation.

8. Evaluation Results

8.1. Participant Summary

The evaluation participants consisted of individuals from different age groups and with varying levels of fitness knowledge and exercise experience. None of the participants were professional, including university students, beginners, and moderately active individuals.

Goal Alignment & Personalization

Overall, participants rated that the recommendations matched their fitness goals with a mean rating of 4.45. However, the raters did not find the results personalized and rather generic. We believe this is caused by the fact that the users were expecting specific exercise routines rather than exercise categories as we proposed. Many of the comments suggested we gave

more personalized unique exercise routines rather than sport categories. Some even suggested we would ask more specific questions and give dietary recommendations as well.

Safety

There were no safety concerns regarding the recommendations the ratings suggest (mean>3).

Overall

Regarding the application UI, the ratings suggest it was easy to use and understandable and the recommendations did make sense according to the information they provided. However, 18.2% of the users did not find the application feasible, meaning they would not prefer to use the application again. This result was partially expected, as several participants perceived the recommendations as insufficiently personalized. Some users also reported that the application did not provide enough guidance or detail to help them construct a complete and practical workout plan. (18.2% of the users)

Quantitative Results

Statements	Construct	Mean	Standard Deviation	Median
S1	Goal Alignment	4.45	0.69	5.0
S2	Personalization	3.64	1.21	4.0
S3	Safety	4.45	0.82	5.0
S4	Utility for Planning	3.64	1.12	4.0

S5	Usability	4.91	0.30	5.0
S6	Retention Intent	3.73	1.35	4.0
S7	Coherence	4.55	1.04	5.0

The evaluation results indicated that most users considered the generated recommendations understandable and generally appropriate for their exercise goals and fitness level. However, they did not feel it was personalized enough.

Qualitative Feedback Analysis

Open-ended responses were analyzed using thematic analysis. Five primary themes emerged from participant feedback.

Theme 1: Desire for Specific Exercise Recommendations

"Daha spesifik egzersizler olabilir" – "More specific exercises could be included"

"The recommended exercises are too generic. It would be much better to divide the recommendations into more categories."

"More specific body shape goals and their corresponding exercises and target muscles can be added."

Theme 2: Request for Explanatory Content

Participants requested explanatory materials to contextualize recommendations:

"Maybe I'd even include some pictures of the exercises. I could also show an example workout plan."

"I would maybe add some resources regarding the exercises recommended, like YouTube videos or websites that explain the exercises."

"Explain the exercises I recommend instead of just listing them."

Theme 3: Desire for Additional Input Variables

Several participants suggested incorporating additional user information:

"I would suggest that any medical issues should be asked as well."

"I would want to be able to enter more information about myself like any special conditions I have that affects my routine or any diet I'm following."

"I'd ask for more detailed information: like your body measurements, body fat percentage."

Theme 4: Scale Interpretation and Clarification

One participant noted ambiguity in the slider scale interpretation:

*"The sliders being from 1-10 is a bit misleading. It could be more clear if it started from 0, because for example when I was entering weight loss goal I said 1 but in reality I do not want to lose weight at all."

Theme 5: Usability Appreciation

Despite suggestions for improvement, participants consistently praised the interface:

"Easy to use and nice to look at in general."

"I really like the interface and how practical and easy to use it."

8.4. Summary of Key Findings

Finding 1: Strong Usability and Goal Alignment

The system achieved high scores for usability (mean = 4.91) and goal alignment (mean = 4.45), indicating successful implementation of core functionality. The convergence properties of the FCM (5-7 iterations across all test cases) demonstrate computational efficiency.

Finding 2: Personalization and Utility Require Enhancement

Lower scores for personalization (mean = 3.64) and utility for planning (mean = 3.64) identify these as primary areas for improvement. Qualitative feedback consistently requested more specific recommendations and explanatory content.

Finding 3: Safety Perception Is High

Safety scores were uniformly high (mean = 4.45) across BMI categories, suggesting that the system's conservative weighting (negative BMI→HIIT weight) and safety disclaimer successfully address user safety concerns.

Finding 4: Scale Interpretation Ambiguity

Participant feedback revealed that the 1-10 slider scale was ambiguous for representing zero-priority goals. A 0-9 or 0-10 scale with clearer labeling would reduce this ambiguity.

Finding 5: Demand for Expanded Inputs and Outputs

Qualitative analysis revealed consistent demand for additional input variables (medical conditions, body fat percentage, dietary information) and more granular output categories (specific exercises, target muscles, example workouts).

8.5 Limitations

Several methodological limitations should be acknowledged:

Sample Size – The number of participants was eleven. The sample of eleven participants limits statistical power and generalizability of findings. Effect sizes should be interpreted cautiously.

Sampling Method – Convenience sampling from the university community may introduce selection bias and limit external validity.

Self-Reported Fitness Level – Fitness level was self-reported rather than objectively measured, introducing potential reporting bias.

BMI as Sole Body Composition Metric – BMI does not distinguish between fat mass and lean mass, limiting its utility for athletic populations.

Short-Term Evaluation – The evaluation measured immediate post-use perceptions rather than long-term behavior change or recommendation adherence.

No Control Condition – The absence of a control condition (e.g., generic recommendations) precludes direct comparison of FCM-generated recommendations against baseline alternatives.

Single Interaction per Participant – Each participant interacted with the system only once, preventing assessment of within-subject consistency or learning effects.

8.6 Future Work

Based on evaluation findings, the following improvements are recommended for future iterations:

- Add explanation feature providing reasoning for each recommendation
- Expand output categories to include specific exercises and target muscles
- Refine slider scale (0-10 with clearer labels)
- Add medical conditions and body fat percentage as optional inputs
- Embed BMI calculator within interface
- Integrate multimedia resources (video demonstrations)
- Implement personalized workout planning feature

8.7 Conclusion

This project successfully developed and evaluated a Fuzzy Exercise Assistant that generates personalized exercise category recommendations based on user BMI, fitness level, and exercise goals. The system demonstrated strong usability (mean = 4.91) and goal alignment (mean = 4.45) across eleven evaluation participants. Quantitative analysis supported hypotheses regarding goal alignment and the relationship between fitness level and HIIT acceptance. Qualitative feedback identified specific opportunities for improvement, particularly in personalization, recommendation granularity, and explanatory content.

The Fuzzy Cognitive Map approach proved suitable for this domain, enabling the encoding of expert knowledge about exercise suitability in a transparent, interpretable weight matrix. The system converged efficiently (5-7 iterations) and maintained robust performance even in the absence of external libraries through a native fallback implementation.

Future work will focus on addressing the identified limitations: expanding the set of input variables, increasing output granularity, incorporating explanatory content, and conducting a larger-scale evaluation with longitudinal follow-up. Despite current limitations, the Fuzzy Exercise Assistant represents a viable first step toward personalized, fuzzy logic-based exercise recommendation systems.

References

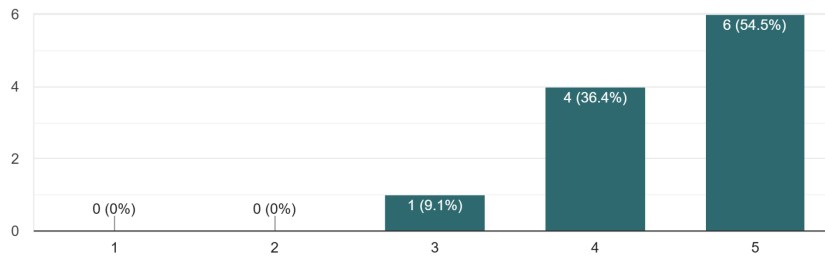
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Appendix

Questionnaire

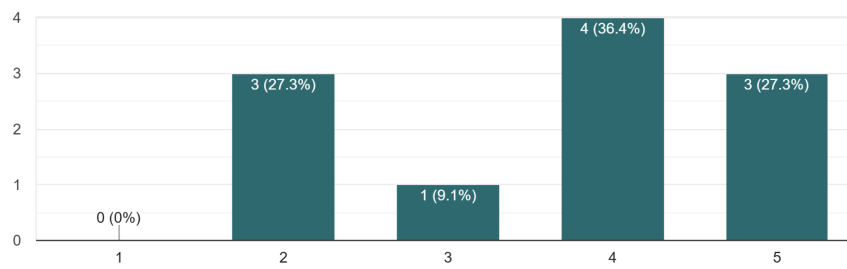
The recommended exercises matched my fitness goals.

11 responses



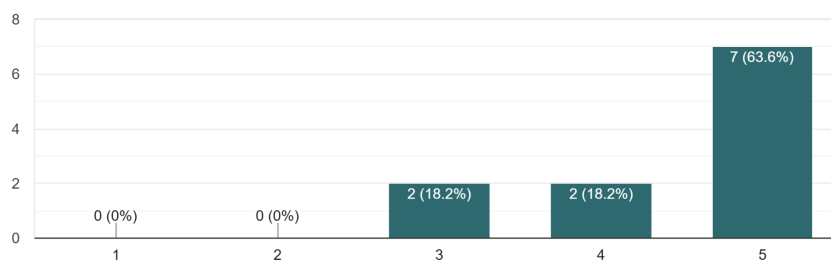
The recommendations felt personalized and not generic.

11 responses



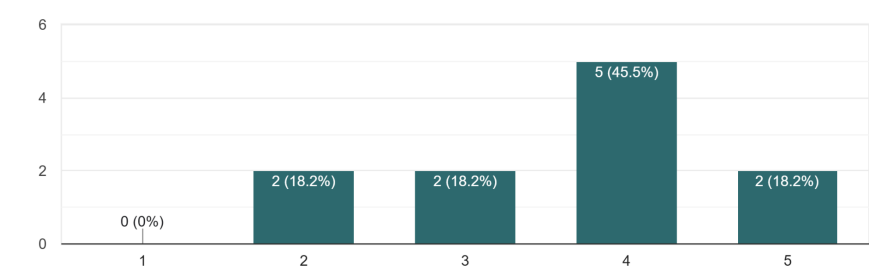
The recommendations seemed safe for me.

11 responses



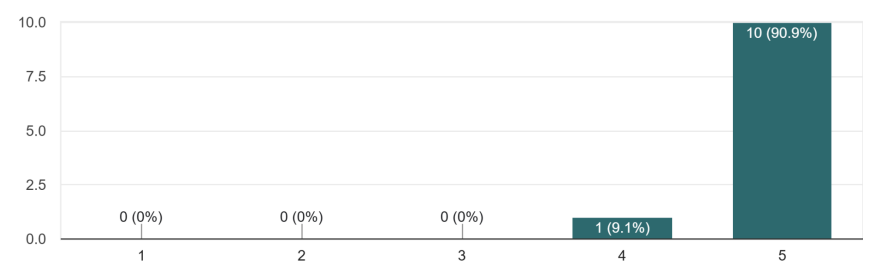
The recommendations were useful for planning my workouts.

11 responses



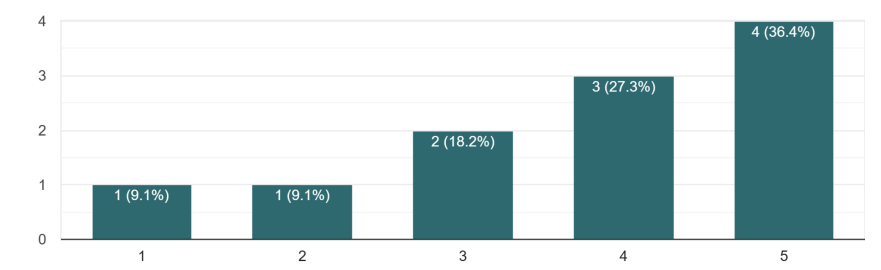
The application was easy to use.

11 responses



I would use this assistant again.

11 responses



The recommendations made sense given the information I entered.

11 responses

